Potentials and Fields

Problem 10.1

Show that the differential equations for V and $\boldsymbol{A}$

$$\nabla^2 V + \frac{\partial}{\partial t}(\nabla \cdot \boldsymbol{A}) = -\frac{1}{\epsilon_0}\rho$$

$$\left(\nabla^2 \boldsymbol{A} - \mu_0\epsilon_0\frac{\partial^2 \boldsymbol{A}}{\partial t^2}\right) - \nabla\left(\nabla \cdot \boldsymbol{A} + \mu_0\epsilon_0\frac{\partial V}{\partial t}\right) = -\mu_0\boldsymbol{J}$$

can be written in the more symmetrical form

$$\Box^2 V + \frac{\partial L}{\partial t} = -\frac{1}{\epsilon_0}\rho$$

$$\Box^2 \boldsymbol{A} - \nabla L = -\mu_0\boldsymbol{J}$$

where

$$\Box^2 \equiv \nabla^2 - \mu_0\epsilon_0\frac{\partial^2}{\partial t^2} \quad \text{and} \quad L \equiv \nabla \cdot \boldsymbol{A} + \mu_0\epsilon_0\frac{\partial V}{\partial t}$$

Solution:

Put the definitions of $\Box^2$ and L into the equations for V:

$$\left(\nabla^2 - \mu_0\epsilon_0\frac{\partial^2}{\partial t^2}\right)V + \frac{\partial}{\partial t}\left(\nabla \cdot \boldsymbol{A} + \mu_0\epsilon_0\frac{\partial V}{\partial t}\right) =$$

$$= \nabla^2 V - \mu_0\epsilon_0\frac{\partial^2 V}{\partial t^2} + \frac{\partial(\nabla \cdot \boldsymbol{A})}{\partial t} + \mu_0\epsilon_0\frac{\partial^2 V}{\partial t^2} =$$

$$= \nabla^2 V + \frac{\partial}{\partial t}(\nabla \cdot \boldsymbol{A}) = -\frac{1}{\epsilon_0}\rho \;\#$$

and for $\boldsymbol{A}$:

$$\left(\boldsymbol{\nabla}^2 - \mu_0\epsilon_0\frac{\partial^2}{\partial t^2}\right)\boldsymbol{A} - \boldsymbol{\nabla}\left(\boldsymbol{\nabla}\cdot\boldsymbol{A} + \mu_0\epsilon_0\frac{\partial V}{\partial t}\right) =$$

$$= \left(\boldsymbol{\nabla}^2\boldsymbol{A} - \mu_0\epsilon_0\frac{\partial^2 \boldsymbol{A}}{\partial t^2}\right) - \boldsymbol{\nabla}\left(\boldsymbol{\nabla}\cdot\boldsymbol{A} + \mu_0\epsilon_0\frac{\partial V}{\partial t}\right) = -\mu_0\boldsymbol{J} \ \#$$

□

Problem 10.2

Problem 10.3

(a) Find the fields, and the charge and current distributions, corresponding to

$$V(\boldsymbol{r},t) = 0, \quad \boldsymbol{A}(\boldsymbol{r},t) = -\frac{1}{4\pi\epsilon_0}\frac{qt}{r^2}\hat{\mathbf{r}}$$

(b) Use the gauge function $\lambda = -(1/4\pi\epsilon_0)(qt/r)$ to transform the potentials, and comment on the result.

Solution:

The **E** and **B** fields are

$$\mathbf{E} = -\boldsymbol{\nabla} V - \frac{\partial \mathbf{A}}{\partial t}$$
$$\mathbf{B} = \boldsymbol{\nabla}\times\mathbf{A}$$

The derivative of **A** with respect to time is

$$\frac{\partial \mathbf{A}}{\partial t} = -\frac{1}{4\pi\epsilon_0}\frac{q}{r^2}\hat{\mathbf{r}},$$

while the curl of **A** in spherical coordinates is

$$\frac{1}{r\sin\theta}\left[\frac{\partial}{\partial\theta}(\sin\theta A_\phi) - \frac{\partial A_\theta}{\partial\phi}\right]\hat{\mathbf{r}} + \frac{1}{r}\left[\frac{1}{\sin\theta}\frac{\partial A_r}{\partial\phi} - \frac{\partial}{\partial r}(rv_\phi)\right]\hat{\boldsymbol{\theta}} + \frac{1}{r}\left[\frac{\partial}{\partial r}(rA_\theta) - \frac{\partial A_r}{\partial\theta}\right]\hat{\boldsymbol{\phi}}.$$

Since $A_\phi = A_\theta = 0$, and $\frac{\partial A_r}{\partial\phi} = \frac{\partial A_r}{\partial\theta} = 0$, the curl of **A** is zero.

(a)

$$\boxed{\begin{aligned} \mathbf{E} &= -\frac{\partial \mathbf{A}}{\partial t} = \frac{1}{4\pi\epsilon_0}\frac{q}{r^2}\hat{\mathbf{r}} \\ \mathbf{B} &= \nabla \times \mathbf{A} = 0 \end{aligned}}$$

(b) Using the gauge function, V and $\mathbf{A}$ are transformed into

$$V'(\mathbf{r},t) = V(\mathbf{r},t) - \frac{\partial \lambda}{\partial t}$$

$$\mathbf{A}'(\mathbf{r},t) = \mathbf{A}(\mathbf{r},t) + \nabla\lambda$$

Writing out the derivative and gradient of λ

$$\frac{\partial \lambda}{\partial t} = -\frac{1}{4\pi\epsilon_0}\frac{q}{r}$$

$$\nabla\lambda = \frac{1}{4\pi\epsilon_0}\frac{qt}{r^2}\hat{\mathbf{r}}$$

gives the transformed potentials

$$V'(\mathbf{r},t) = \frac{1}{4\pi\epsilon_0}\frac{q}{r}$$

$$\mathbf{A}'(\mathbf{r},t) = -\frac{1}{4\pi\epsilon_0}\frac{qt}{r^2}\hat{\mathbf{r}} + \frac{1}{4\pi\epsilon_0}\frac{qt}{r^2} = 0$$

$\mathbf{E}$ is now given by $-\nabla V$ (instead of by $-\frac{\partial \mathbf{A}}{\partial t}$), while $\mathbf{B}$ is again 0, this time directly because $\nabla \times \mathbf{A} = \nabla \times 0 = 0$.

$$\boxed{\begin{aligned} \mathbf{E} &= -\nabla V = -\frac{1}{4\pi\epsilon_0}\frac{q}{r^2}\hat{\mathbf{r}} \\ \mathbf{B} &= \nabla \times \mathbf{A} = 0 \end{aligned}}$$

The result is exactly the same as in (a).

□

Problem 10.4

Suppose $V = 0$ and $\mathbf{A} = A_0 \sin(kx - \omega t)\hat{\mathbf{y}}$, where $A_0, \omega,$ and k are constants. Find $\mathbf{E}$ and $\mathbf{B}$, and check that they satisfy Maxwell's equations in vacuum. What conditions must you impose on ω and k?

Solution:

Maxwell's equations in vacuum are

$$\text{(i)}\nabla\cdot\mathbf{E}=\frac{1}{\epsilon_0}\rho \qquad \text{(iii)}\nabla\times\mathbf{E}=-\frac{\partial\mathbf{B}}{\partial t}$$

$$\text{(ii)}\nabla\cdot\mathbf{B}=0 \qquad \text{(iv)}\nabla\times\mathbf{B}=\mu_0\mathbf{J}+\mu_0\epsilon_0\frac{\partial\mathbf{E}}{\partial t}$$

where as before we set

$$\mathbf{E}=-\nabla V-\frac{\partial\mathbf{A}}{\partial t}$$
$$\mathbf{B}=\nabla\times\mathbf{A}$$

This gives

$$\mathbf{E}=-\frac{\partial\mathbf{A}}{\partial t}=-\omega A_0\cos(kx-\omega t)\hat{\mathbf{y}}$$
$$\mathbf{B}=\frac{\partial A_y}{\partial x}\hat{\mathbf{z}}=kA_0\cos(kx-\omega t)\hat{\mathbf{z}}$$

Since $\rho=0$, this directly gives $\nabla\cdot\mathbf{E}=0$. (ii) gives $\nabla\cdot\mathbf{B}=0$.

From (iv) we have the relation, since the current density $\mathbf{J}=0$,

$$\nabla\times\mathbf{B}=\mu_0\epsilon_0\frac{\partial\mathbf{E}}{\partial t}$$
$$\downarrow$$
$$\nabla\times\mathbf{B}=-\frac{\partial B_z}{\partial x}\hat{\mathbf{y}}=k^2A_0\sin(kx-\omega t)\hat{\mathbf{y}}$$
$$\mu_0\epsilon_0\frac{\partial\mathbf{E}}{\partial t}=\mu_0\epsilon_0\omega^2A_0\sin(kx-\omega t)\hat{\mathbf{y}}$$

From this we see that $k^2=\mu_0\epsilon_0\omega^2\rightarrow k=\sqrt{\mu_0\epsilon_0}\omega\rightarrow k=\frac{\omega}{c}$

From (iii) we have the relation

$$\nabla\times\mathbf{E}=-\frac{\partial\mathbf{B}}{\partial t}$$
$$\downarrow$$
$$\nabla\times\mathbf{E}=\frac{\partial E_y}{\partial x}\hat{\mathbf{z}}=\omega kA_0\sin(kx-\omega t)\hat{\mathbf{z}}$$
$$-\frac{\partial\mathbf{B}}{\partial t}=\omega kA_0\sin(kx-\omega t)\hat{\mathbf{z}}$$

Equations (i)-(iv) are all verified with the condition $k=\frac{\omega}{c}$.

□

Problem 10.5

Which of the potentials

$$\text{(i) } V = 0, \quad \mathbf{A} = \begin{cases} \frac{\mu_0 k}{4c}(ct - |x|)^2 \hat{\mathbf{z}}, & \text{for } |x| < ct \\ 0 & \text{for } |x| > ct \end{cases}$$

$$\text{(ii) } V = 0, \quad \mathbf{A} = -\frac{1}{4\pi\epsilon_0}\frac{qt}{r^2}\hat{\mathbf{r}}$$

$$\text{(iii) } V = 0, \quad \mathbf{A} = A_0 \sin(kx - \omega t)\hat{\mathbf{y}}$$

are in the Coulomb gauge? Which are in the Lorenz gauge?

Solution:

The Coulomb gauge is

$$\boldsymbol{\nabla} \cdot \mathbf{A} = 0$$

and the Lorenz gauge is

$$\boldsymbol{\nabla} \cdot \mathbf{A} = -\mu_0 \epsilon_0 \frac{\partial V}{\partial t}$$

Starting with checking the Coulomb gauge, we get

$$\text{(i) } \boldsymbol{\nabla} \cdot \mathbf{A} = \boldsymbol{\nabla} \cdot \left(\frac{\mu_0 k}{4c}(ct - |x|)^2 \hat{\mathbf{z}} \right) = 0$$

$$\text{(ii) } \boldsymbol{\nabla} \cdot \mathbf{A} = \boldsymbol{\nabla} \cdot \left(-\frac{1}{4\pi\epsilon_0}\frac{qt}{r^2}\hat{\mathbf{r}} \right) = -\frac{qt}{4\pi\epsilon_0} \boldsymbol{\nabla} \cdot \left(\frac{\hat{\mathbf{r}}}{r^2} \right) = -\frac{qt}{4\pi\epsilon_0}\delta^3(\mathbf{r})$$

$$\text{(iii) } \boldsymbol{\nabla} \cdot \mathbf{A} = \boldsymbol{\nabla} \cdot (A_0 \sin(kx - \omega t)\hat{\mathbf{y}}) = 0$$

(i) and (iii) are in the Coulomb gauge, but not (ii). Next, checking Lorenz gauge we see that the divergence of $\mathbf{A}$ should be proportional to the time derivative of V, but since $V = 0$ for (i) and (iii), they are also in the Lorenz gauge. This does not apply to (ii) since the divergence of $\mathbf{A} \neq 0$ while $\frac{\partial V}{\partial t} = 0$.

(i) Both Coulomb and Lorenz.

(ii) Neither Coulomb or Lorenz.

(iii) Both Coulomb and Lorenz. □

Problem 10.6

Problem 10.7

Problem 10.8

Problem 10.9

Problem 10.10

Problem 10.11

Problem 10.12

Problem 10.13

Problem 10.14

Problem 10.15

Problem 10.16

Problem 10.17

Problem 10.18

Problem 10.19

Problem 10.20

Problem 10.21

Problem 10.22

Problem 10.23

Problem 10.24

Problem 10.25

Problem 10.26

Problem 10.27

Problem 10.28

Problem 10.29

Problem 10.30

Problem 10.31

Problem 10.32

Problem 10.33

Problem 10.34

Problem 10.35

Problem 10.36

Problem 10.37

Problem 10.38